

Shoumik Saha

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 [Google Scholar](https://scholar.google.com/citations?user=smksaha)

RESEARCH INTEREST

- **Machine Learning** - Security & Reliability of GenAI, LLM Alignment & Hallucination
- **Computer Security** - Adversarial Attacks & Defenses, Malware

EXPERIENCE

Applied Scientist Intern

Amazon AWS

Managers - [Varun Kumar](#), [Zijian Wang](#)

May 2025 – Aug 2025

Full Time

Graduate Research Assistant

Department of Computer Science

[University of Maryland](#)

Supervisor - [Dr. Soheil Feizi](#)

Aug 2023 – Present

Full Time

Applied Scientist Intern

Amazon AWS

Manager - [Zijian Wang](#)

Jun 2024 – Aug 2024

Full Time

Graduate Research Assistant

Maryland Cybersecurity Center

Supervisor - [Dr. Tudor Dumitras](#)

Aug 2022 – Jul 2023

Full Time

Lecturer

Department of Computer Science and Engineering

United International University

Jul 2021 – Jul 2022

Full Time

Research Assistant

Data Science and Engineering Research Lab

Bangladesh University of Engineering and Technology (BUET)

Mar 2021 – Jul 2022

Part Time

EDUCATION

University of Maryland - College Park (UMD)

Ph.D. in Computer Science (Ongoing)

Advisor – [Dr. Soheil Feizi](#)

Aug 2022 – Present

CGPA – 3.814/4.0

University of Maryland - College Park (UMD)

M.S. in Computer Science

Advisor – [Dr. Soheil Feizi](#)

Aug 2022 – Aug 2024

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering

Thesis supervisor – [Dr. Atif Hasan Rahman](#)

Mar 2016 – Feb 2021

Advanced GPA – 3.79/4.0

PUBLICATIONS

- **Breaking the Code: Security Assessment of AI Code Agents Through Systematic Jailbreaking Attacks**
[Shoumik Saha, J Chen*, S Mayers*, SK Gouda, Z Wang, V Kumar]
[Under Review](#)([arXiv link](#))
- **Almost AI, Almost Human: The Challenge of Detecting AI-Polished Writing**
[Shoumik Saha, S Feizi]
[ACL 2025](#) ([Association for Computational Linguistics](#)) ([Paper Link](#))
Media Coverage: [The New York Times](#), [Plagiarism Today](#), [The Science Archive](#), [The Cheat Sheet](#)

- **Adversarial Paraphrasing: A Universal Attack for Humanizing AI-Generated Text**
[Y Cheng, VS Sadasivan, **Shoumik Saha***, M Saberi*, S Feizi]
NeurIPS 2025 (Conference on Neural Information Processing Systems) ([Paper Link](#))
- **ProcVQA: Benchmarking the Effects of Structural Properties in Mined Process Visualizations on Vision–Language Model Performance**
[KT Zinat*, SM Abrar*, **Shoumik Saha**, S Duppala, SN Sakhamuri, Z Liu]
EMNLP 2025 (Empirical Methods in Natural Language Processing) ([Paper Link](#))
- **LLM-Check: Investigating Detection of Hallucinations in Large Language Models**
[G Sriramanan, S Bharti, VS Sadasivan, **Shoumik Saha**, P Kattakinda, S Feizi]
NeurIPS 2024 (Conference on Neural Information Processing Systems) ([Paper Link](#))
- **Fast Adversarial Attacks on Language Models In One GPU Minute**
[VS Sadasivan, **Shoumik Saha***, G Sriramanan*, P Kattakinda, A Chegini, S Feizi]
ICML 2024 (International Conference on Machine Learning) ([Paper Link](#))
Media Coverage: [The Register](#)
- **DRSM: De-Randomized Smoothing on Malware Classifier Providing Certified Robustness**
[**Shoumik Saha**, W Wang, Y Kaya, S Feizi, T Dumitras]
ICLR 2024 (International Conference on Learning Representations) ([Paper Link](#))
- **MAlign: Explainable Static Raw-byte Based Malware Family Classification using Sequence Alignment**
[**Shoumik Saha**, S Afroz, A Rahman]
Computers & Security Journal ([Paper Link](#))
- **ML-Based Behavioral Malware Detection Is Far From a Solved Problem**
[Y Kaya, Y Chen, Marcus Botacin, **Shoumik Saha**, F Pierazzi, L Cavallaro, D Wagner, T Dumitras]
SaTML 2025 (Secure & Trustworthy Machine Learning) ([Paper Link](#))
- **Contrastive Self-Supervised Learning Based Approach for Patient Similarity: A Case Study on Atrial Fibrillation Detection from PPG Signal**
[**Shoumik Saha***, S Shanto*, A Rahman, M Masud, ME Ali]
([arXiv Link](#))

SCHOLARSHIPS & ACHIEVEMENTS

- Awarded 'Dean's Fellowship' from the University of Maryland
- Awarded 'Innovation Fund' for research from the ICT division of Bangladesh government
- Achieved 'Merit Stipends' from BUET in five out of seven terms
- Awarded 'Dean's Award' in Junior year for extra-ordinary result
- Achieved 'Talent-Pool Scholarship' in High School

TEACHING

Teaching Assistant

University of Maryland

Fall 2023

Data 200: Knowledge in Society: Science, Data and Ethics

Lecturer

United International University

Fall 2021 - Summer 2022

Discrete Mathematics, Data Structure and Algorithm, Operating Systems

SERVICES

- **Reviewer**, top-tier ML conferences such as ICML 2024, ICLR 2025, NEURIPS 2025, etc.
- **External Reviewer**, top-tier Computer Security conference, including USENIX 2023
- **Lab Instructor**, United International University
- **General Secretary**, BUET Photographic Society